

The Path from User Application to Hardware Access

The Kernel is the connection between the computer hardware and end user applications. The levels of abstraction increase, the requirement for absolute control is relinquished to the Kernel and its device drivers to perform the desired actions. For example, to read a file, we could use the `cat` command provided by the shell without having to know exactly how to select addresses and buses, or how to reserve memory to place the information into it. The Kernel provides resource management and services interfaces.

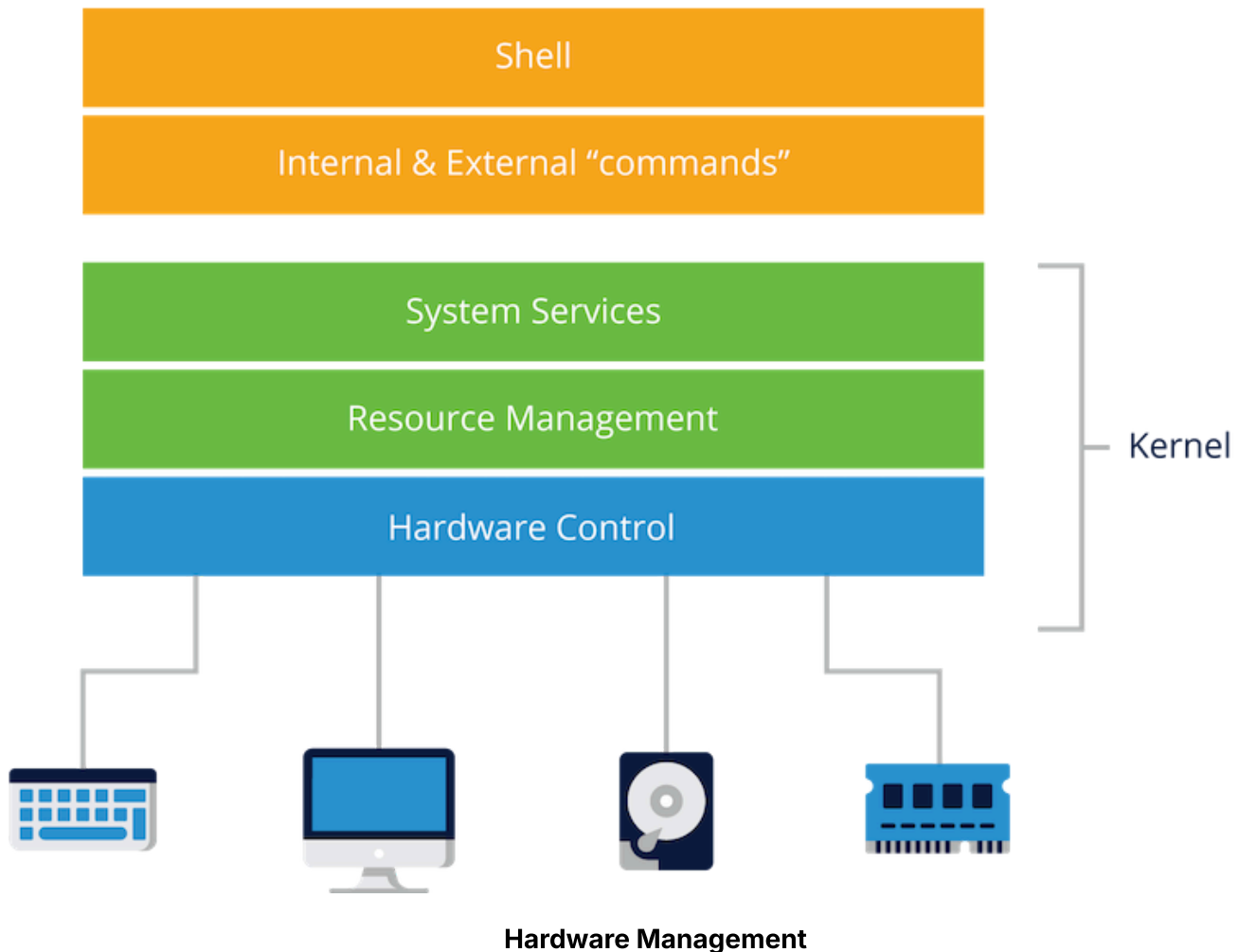
Stepping back, we can see the relationship more clearly:



Connecting to Hardware (x86)

Applications are programs that perform specific activities. In other words, applications assist the user in completing a task such as word processing, web browsing, sending emails, instant messaging, gaming, and system administration utilities.

The application program uses the operating system (OS). The OS controls the hardware, so when we use the computer, the application needs to talk to the OS. In order to manage the input and output devices, the OS needs to perform its control functions. As end-users, we interact with input devices such as typing. Typing a letter or moving the mouse and performing other functions is interpreted by the OS, the information is passed on to the application, and the application then performs the specific tasks that we want to conduct.



In the preceding image, the computer software has been divided into some common components, both for the operating system and the user interface. Notice that the hardware is directly connected to the kernel, which provides services to the user. This interface is known as *device drivers*, rather than *hardware control*, since there are many devices to control. Some of the device drivers act on software "devices", which programs that react like real hardware, but are made entirely out of software.